

PRITHVI KRISHNASWAMY

(360) 990-5995 | 1811prithvi@gmail.com | LinkedIn: [Prithvi-Krishnaswamy](#) | Website: <https://krishpri-1811.github.io/>

Engineering Skills: Altium Designer, LTSpice, Python, Cadence Virtuoso/Spectre, MATLAB, Simulink, PLECS, Ansys HFSS

Education Credentials

Bachelor of Science: Electrical Engineering | **University of Washington (UW)**, Seattle, WA | Graduation: 06/2025
Minor: Applied Mathematics | **Coursework: Circuit Theory, Energy Systems, Applied Electromagnetics** | **GPA: 3.57**
Master of Science: Electrical Engineering | **University of Washington (UW)**, Seattle, WA | Graduation: 06/2027
Coursework: Power Electronics, Control Systems, Analog Integrated Circuits, Antenna Design | **GPA: 3.82**

Professional Experience

Avionics Hardware Design Engineering Intern (Incoming) | **Stoke Space** | Kent, WA | Sep 2026 – Dec 2026

- Design & Testing Servo Controllers For [Nova](#): Stoke Space's fully reusable launch vehicle!

EV Electrical Design Engineering Intern | **Rivian/Volkswagen Technologies** | Irvine, CA | Jun 2026 – Sep 2026

- Tuned reliability of **Pierce Oscillator Circuit** across **10 Zonal Controllers**, **4 vehicle families** & up to **~100,000 vehicles**.
- Characterized **12V/48V HSDs** & validated **RCP Emulation** for Zonal-Controlled edge nodes, advancing a SW-less architecture **eliminating local MCUs**, device-resident application software/OTA burden & reduced wiring complexity
- Normal-Load Efficiency/Thermal Performance** characterization of a **2-Phase 48V-to-15V GaN DC-DC Converter** across 1-34 A loads, 36V/48V/54V inputs, and -40 C to 85 C test corners. Demonstrated **98.03% efficiency**
- Tested & Updated a **48V-to-12V short-circuit protection** test board for **12V HSDs** against **440 A transients** under a **48 V Bus Fault**. Compared component-level surge-energy, clamping voltage and recovery performance

Aircraft Electrical Design Engineering Intern | **Shield AI** | Dallas, TX | Jan 2026 – Jun 2026

- Hardware Bring-up, DVT & Qualification of V-BAT [ViDAR](#) eXtensible Processing Module (XPM). Validated **Electrical, Thermal, OS & Mission Software** Parameters for 1st stage Experimental Flight Test (XFT)
- Prototyped DAQ Interposer PCB to validate connector stability. Scripted Python vibration tests & logged data using **LabVIEW**
- Owned EE responsibilities for V-BAT **Battery Management System (BMS)** enabling cell-level Temperature, Voltage, State of Health (SOH), and State of Charge (SOC) monitoring. Added enhanced Protection and Sequence Mechanisms. Qualified Thermal, Vibration & Quiescent Current Discharge Performance to match design/operating parameters

Datacenter Hardware Design Engineering Intern | **Nvidia** | Durham, NC | Jun 2024 – Sep 2024

- Designed a simplified GPU-emulator PCB in **Cadence Allegro** that emulated key power, signal, and I²C interfaces of [the GB200 server switch tray](#), **cutting test-fixture expenses by $\approx 85\%$** and accelerating bring-up cycles.
- Qualified **Single/Multi Phase Buck & LDO** rails on [50xx GPU](#) SLT boards. Characterized **FRA Bode Plot, Transient, Efficiency, Input Droop, Output Ripple**, and AMR measurements to achieve **$\geq 55^\circ$ phase margin** and **< 30 mVpp** at rated load.
- Re-architected a Power Adapter PCB for 100+ engineers— transitioned to CEM5 high-current connectors, added OVP/UVP, inrush control - to improve safe & flexible AC/DC and DC/DC power-conversion for DGX/MGX server platforms.

Plasma Etch Silicon Engineering Intern | [Intel](#) | Hillsboro, OR | Jan 2023 – Dec 2023

- Innovated a custom **Deep Reactive Ion Etching (DRIE)** [Bosch Process](#), achieving **50:1 to 100:1 aspect ratio** trenches and holes for MEMs Through-Silicon Vias (TSVs). **Enhanced etching precision by 3x** baseline on 20 validated silicon wafers.

Automotive Technician & App Developer | **Singapore Armed Forces** | Singapore | Oct 2019 – July 2021

- Mechanical & Electrical maintenance on 4 different fleets of 20 vehicles. Conducted Failure Analysis, debugging & validation of Batteries, Filters & Drivetrains. Catalogued defects & ordered necessary replacements.
- Constructed a user-friendly front-end interface for a multiplatform application aimed at increasing workshop safety awareness; designed back-end connections that ensured secure handling of user input data for over 100 users
- Beta-tested the application with documentations and [training videos](#) that **increased user adoption by a factor of 4**

Projects

Electronics Engineer | [UW Formula Motorsports \(FSAE\)](#) | Seattle, WA | Jul 2025 – Jun 2026

- Architecting an [Integrated Powertrain Module](#) to reduce HV harness length, simplify packaging, and improve EMI behavior
- Spearheading architecture & design of a custom **900W**, isolated [Resonant LLC DC – DC Converter](#) **reducing battery costs by 90%**, while achieving **dual compatibility with 400V & 600V HV pack architectures**
- Renovating Schematic & Layout, in Altium, of a custom **26 kVA, 540V DC** high-performance [Quad – Motor Inverter](#) to **decrease car weight by 11kg** and increase **efficiency up to $\sim 98\%$**

Extracurriculars

Teaching Assistant | **UW Department of Electrical & Computer Engineering** | Seattle, WA | Jun 2023 – Jun 2025

- Built a [line-following robot platform](#) **used by 100+ students**, to teach full-stack prototyping (PCB $\rightarrow$ firmware $\rightarrow$ validation).
- Coordinated 100+ office-hour sessions (≈ 30 students/week) and streamlined grading for labs & HW

Lead Research Assistant | [Washington Nanofabrication Facility](#) | Seattle, WA | May 2022 - Current

- Validated Vision **Reactive Ion Etcher (RIE)** quality & reliability. Subjected ~ 100 silicon wafers across various processes, verifying trench depth & uniformity using a **Scanning Electron Microscope (SEM)**. Published process notes for lab users.
- Doubled etching rate of Oxford **Inductively Coupled Plasma (ICP) Etcher** for **PECVD** & Thermal Oxide deposited Silicon Dioxide wafers. Documented parameters for reproducibility.